Subgroup intersections under central extensions

Human Compression project

Private component for the recovery of L019

Abstract

A central extension can remove an intersection defect only when the corresponding quotient embeds in a quotient of its central kernel. In particular, an odd factor in the intersection index survives every extension by central involutions. This supplies a uniform replacement for several uses of an unrestricted endpoint-untwisting claim in the specified source. An exact check of its eleven finite drawings verifies ten exclusions and identifies a positive exception in the remaining drawing. No conclusion about the source's main stability theorem is claimed here.

1 The difference between two lifts

All groups in the finite applications are actual permutation groups on the displayed point set. A marking $s_0, \dots, s_{r-1}$ is C when, for every two label sets I, J, its generated subgroups satisfy $G_I \cap G_J = G_{I \cap J}$. The result below does not require involutions or the string commutation condition.

Lemma 1. *Let $\pi : \widetilde{G} \to G$ be a surjective homomorphism with central kernel E. Let $\widetilde{U}, \widetilde{V} \le \widetilde{G}$, put $U = \pi(\widetilde{U})$, $V = \pi(\widetilde{V})$ and $K_U = \widetilde{U} \cap E$, $K_V = \widetilde{V} \cap E$. Then $P = \pi(\widetilde{U} \cap \widetilde{V})$ is a normal subgroup of $U \cap V$, and*

$$(U \cap V)/P \;\hookrightarrow\; E/(K_U K_V).$$

Proof. For $x \in U \cap V$, choose lifts $u \in \widetilde{U}$ and $v \in \widetilde{V}$ with image x. Their difference uv^{-1} lies in E. Replacing the two lifts changes this difference by an element of $K_U K_V$, so

$$\delta(x) = uv^{-1} K_U K_V$$

is well defined. Centrality of E makes it abelian and makes the displayed product of subgroups a subgroup. If u_x, v_x, u_y, v_y are the chosen lifts for x, y, the lifts $u_x u_y, v_x v_y$ give

$$u_x u_y (v_x v_y)^{-1} = u_x (u_y v_y^{-1}) v_x^{-1} = (u_y v_y^{-1})(u_x v_x^{-1}).$$

Thus δ is a homomorphism into $E/(K_U K_V)$.

If x has a common lift in the two subgroups, then $\delta(x) = 1$. Conversely, suppose $uv^{-1} = k_U k_V$ with $k_U \in K_U$ and $k_V \in K_V$. The element $k_U^{-1} u = k_V v$ belongs to both subgroups and maps to x. Hence $\ker \delta = P$. The first isomorphism theorem gives the asserted normality and embedding. □

Theorem 2. *Suppose a finite marked group $(\widetilde{G}; \widetilde{s}_i)$ is C and $\pi : \widetilde{G} \to G$ has central kernel E. Write $s_i = \pi(\widetilde{s}_i)$. For every pair I, J of label sets, $G_{I \cap J}$ is normal in $G_I \cap G_J$, and*

$$(G_I \cap G_J)/G_{I \cap J} \hookrightarrow E/(E \cap \widetilde{G}_I)(E \cap \widetilde{G}_J).$$

In particular, if E is a p-group, then $[G_I \cap G_J : G_{I \cap J}]$ is a power of p.

Proof. Apply Lemma 1 to the two marked subgroups upstairs. Their intersection is $\widetilde{G}_{I\cap J}$ by the C-property, and its image is exactly $G_{I\cap J}$ because the homomorphism sends each of its specified generators to the specified old generator. The order of the displayed quotient divides the order of the finite central quotient on the right. □

Corollary 3. *Let s_i be a marking of a finite group G, and let E be a finite elementary abelian 2-group. Replace the marks by (s_i, e_i) in $G \times E$, for arbitrary $e_i \in E$. If an original intersection index has an odd prime divisor, the resulting marking is not C.*

Proof. The subgroup generated by the new marks projects onto G. Its kernel is a subgroup of the central group E. The conclusion follows from Theorem 2 with $p = 2$. □

The corollary permits simultaneous twists at any labels, not merely one endpoint. For a permutation application, disjoint supports for the original action and the commuting extra involutions supply precisely the required direct product. An actual homomorphism to the original group is essential: an arbitrary quotient or an unproved separation of restriction images cannot replace it. The theorem does not justify unrestricted untwisting when the old intersection index is a power of two. In particular it does not undo the previously verified counterexample to Proposition 2.4 of [1].

2 The finite diagrams in the source

The source is the exact author edition arXiv:2212.12723v2, 11 January 2023. Its Proposition 2.5, physical page 7, displays eleven rank-four markings. Number vertices from left to right along the upper row and then from left to right along the lower row, omitting empty diagram positions. The attached numeric data record every transposition of every mark under this convention. Both the primary TeX and the native PDF drawing have been inspected; no inferred abstract isomorphism identifies the inputs.

Put $U = \langle s_1, s_2, s_3 \rangle$, $V = \langle s_0, s_1, s_2 \rangle$, $I = U \cap V$ and $M = \langle s_1, s_2 \rangle$. Exact finite calculations give:

Source diagram	$\lvert U\rvert$	$\lvert V\rvert$	$\lvert I\rvert$	$\lvert M\rvert$	$[I : M]$
1	4320	10080	240	20	12
2	2880	10080	144	24	6
3	720	720	120	10	12
4	4320	10080	240	20	12
5a	10080	10080	480	20	24
5b	10080	10080	240	20	12
6a	5040	1440	240	20	12
6b	5040	1440	120	20	6
7	5040	720	144	24	6
8	5040	720	144	24	6
9	60	720	10	10	1

Consequently the first ten markings fail the intersection property, and so do all their central involution twists covered by Corollary 3. This is stronger than the required individual failures. It supplies a valid way to exclude larger markings containing such a twisted submarking, since every submarking of a C-marking is C. Each use in a larger permutation graph must still identify its actual central extension and its specified four labels.

The source's numerical description of diagram 5b overstates $|I|$ by a factor of two. The exclusion remains valid, with index 12. Diagram 9 requires a different correction: it is a

C-marking, not an exclusion. Its literal permutations are

$$s_0 = (1\ 2), \quad s_1 = (2\ 3)(4\ 5), \quad s_2 = (3\ 4)(5\ 6), \quad s_3 = (4\ 5)(6\ 7).$$

They generate S_7 and have adjacent product orders $6, 5, 5$. The two distant pairs involving s_0 have disjoint supports; the remaining distant pair shares the transposition $(4\ 5)$ and has disjoint remaining transpositions, so the string condition holds. The finite certificate verifies every one of the 256 subset intersections, including the empty subset. Thus the claimed universal exclusion in Proposition 2.5 is false for the displayed diagram 9. Its reported group $U \cong A_6$ and intersection $I \cong A_5$ have orders 360 and 60; the literal drawing instead gives orders 60 and 10.

3 Finite verification and remaining scope

The finite verification uses the following complete procedure. Encode a permutation by its ordered list of images. Starting with the identity, repeatedly right-multiply each newly found permutation by each specified generator and retain every previously unseen product. At termination the retained set contains the generators, consists of their words, and is closed under multiplication by them. It is therefore exactly the generated group. For the table, perform this procedure for U, V, M and intersect the two literal sets for U, V. Their orders never exceed 10080. For diagram 9, perform it for all sixteen label subsets and test all 256 intersections. The full group has 5040 elements on seven points, proving that it is S_7. This is a finite certificate for the specified inputs, not a claim about unenumerated degrees or arbitrary markings.

The implementation validates the input matchings, nonidentity and distinctness of all four marks, the string commutations and every displayed order. Optimized Python is rejected. A separate GAP calculation checks the same literal permutation inputs using its subgroup algorithms. The uniform statement about all central twists follows from the written proof in the first section, not from sampling a few twists.

The finite exclusions and their use as local submarkings come from [1, Proposition 2.5]. The difference homomorphism provides the present repair of the untwisting interface; no historical priority claim is made for this elementary argument. The corrected entries are source-audit findings, not answers to open conjectures. The high-rank existence of suitable splits, admissible rank extensions, inverse contraction, and the global counting bijection of L019 remain separate proof obligations. In particular, uses of the excluded diagram 9 in the source's subsequent attachment argument must be checked again. No complete original recovery or refutation of the main stability theorem is asserted.

References

[1] P. J. Cameron, M. E. Fernandes and D. Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2, 11 January 2023, Proposition 2.5 and physical page 7.